# CardioVision AI

Multi-Vessel Risk Assessment & Clinical Vector Evaluation

**RESEARCH PROTOTYPE EVALUATION REPORT — NOT FOR CLINICAL DIAGNOSIS**

**Evaluation Date:** Wed, Oct 14, 2026, 06:56 AM
**Patient ID:** CV-DEMO-001
**Architecture:** 57-Feature Four-Model Architecture
**Active Mode:** Standard Production Mode

| PATIENT NAME | PATIENT ID | AGE / BIOLOGICAL SEX | HEIGHT / WEIGHT / BMI |
|---|---|---|---|
| Alex Morgan | CV-DEMO-001 | 58 yrs · Male | 170 cm / 72 kg (24.9 kg/m²) |

## 1. Cardiovascular Prediction Summary

Multi-vessel predictive classification derived from the 57-dimensional clinical feature vector.

| OVERALL CAD | LAD ARTERY | LCX ARTERY | RCA ARTERY |
|---|---|---|---|
| **91.7%** | **79.0%** | **65.0%** | **51.7%** |
| CAD | Stenotic | Stenotic | Stenotic |
| cad_model.pkl | lad_model.pkl | lcx_model.pkl | rca_model.pkl |
| Threshold: 50.0% | Threshold: 50.0% | Threshold: 50.0% | Threshold: 50.0% |
| High Est. Prob (≥70%) | High Est. Prob (≥70%) | Moderate Est. Prob (50–69%) | Moderate Est. Prob (50–69%) |

**Model Estimation Notice:** Values are model-estimated statistical classification probabilities, not clinically calibrated epidemiological risk scores or measured anatomical luminal stenosis narrowing percentages.

## 2. Clinical Input Measurements Summary

Patient clinical parameters from active evaluation session evaluated against informational reference benchmarks.

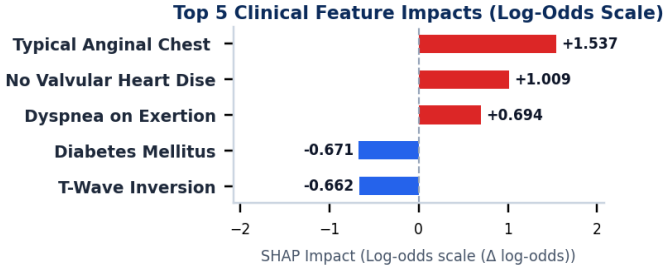
| Clinical Parameter | Patient Value | General Reference Benchmark | Informational Evaluation Status |
|---|---|---|---|
| **Patient Stature (Height)** | **170 cm** | 50 – 250 cm (Physiological reference) | Within adult stature reference |
| **Body Mass Index (BMI)** | **24.9 kg/m²** | 18.5 – 24.9 kg/m² (WHO standard) | Normal weight reference (18.5 – 24.9 kg/m²) |
| **Resting Systolic BP (BP)** | **130 mmHg** | < 120 mmHg (resting systolic) | Above reference (≥ 130 mmHg, Informational) |
| **Resting Pulse Rate (PR)** | **70 bpm** | 60 – 100 bpm | Within standard resting range |
| **Left Ventricular EF (EF-TTE)** | **50%** | ≥ 50% | LVEF ≥ 50% (Standard systolic function) |
| **Fasting Blood Sugar (FBS)** | **98 mg/dL** | 70 – 99 mg/dL | Within fasting range (70–99 mg/dL) |
| **Serum Creatinine (CR)** | **1.0 mg/dL** | 0.7 – 1.3 mg/dL | Within displayed adult reference range (Informational) |
| **LDL Cholesterol** | **100 mg/dL** | < 100 mg/dL | Above standard reference (≥ 100 mg/dL, Informational) |
| **HDL Cholesterol** | **39 mg/dL** | > 40 mg/dL (M) / > 50 mg/dL (F) | Below reference (< 40 mg/dL, Informational) |
| **Triglycerides (TG)** | **122 mg/dL** | < 150 mg/dL | Within reference (< 150 mg/dL) |
| **Evaluated Clinical Vector** | **57 Model Inputs** | Standardized 7-domain vector | 34 numeric + 23 binary clinical indicators synchronized |

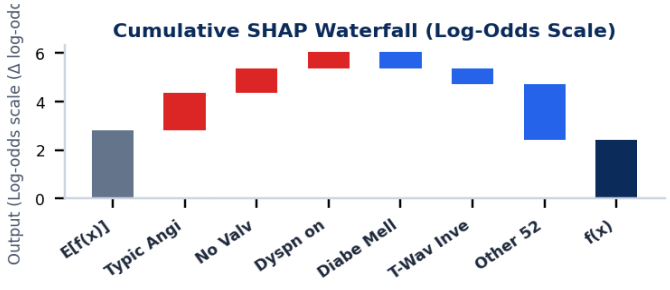
## 3. Patient-Specific SHAP Explainability

Feature contributions calculated via game-theoretic Shapley values. **Scale Note:** TreeExplainer outputs additive probability contributions (Δ probability). LinearExplainer outputs additive log-odds contributions (Δ log-odds). Scales are strictly separated.

**CAD Model Explainability** (cad_model.pkl) · Method: **LinearExplainer (Linear SHAP)** · Target: **CAD** · **[LOG-ODDS SCALE]**
Expected Baseline E[f(x)]: **2.810 log-odds** | Final Patient Output f(x): **2.407 log-odds** (σ(f(x)) = **91.7%** est. prob) | **✓ Additivity Verified: E[f(x)] + Σ ϕ_i = f(x)**

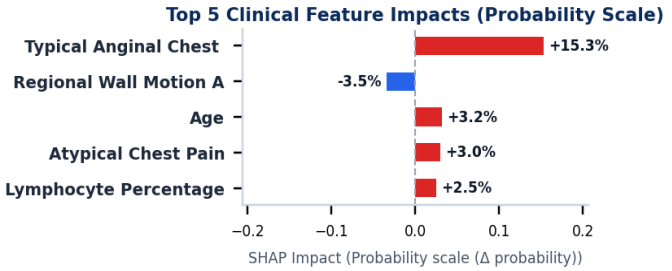
| Rank | Clinical Feature | Category | Patient Value | SHAP (log-odds) | Impact Direction |
|---|---|---|---|---|---|
| #1 | Typical Anginal Chest Pain | Symptoms | Present (1) | +1.537 log-odds | Increases Risk |
| #2 | No Valvular Heart Disease | Valvular Disease | Present (1) | +1.009 log-odds | Increases Risk |
| #3 | Dyspnea on Exertion | Symptoms | Absent (0) | +0.694 log-odds | Increases Risk |
| #4 | Diabetes Mellitus | Medical History | Absent (0) | -0.671 log-odds | Decreases Risk |
| #5 | T-Wave Inversion | ECG & Telemetry | Absent (0) | -0.662 log-odds | Decreases Risk |

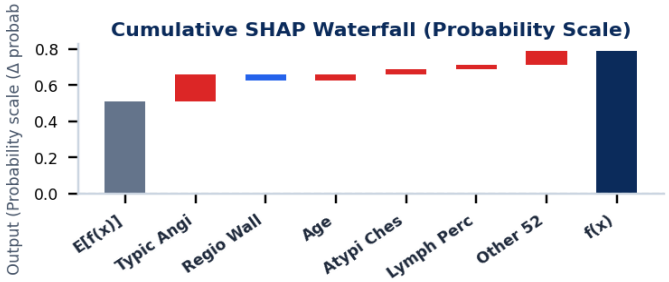


**LAD Artery Model Explainability** (lad_model.pkl) · Method: **TreeExplainer (Tree SHAP)** · Target: **Stenotic** · **[PROBABILITY SCALE]**
Expected Baseline E[f(x)]: **50.7%** | Final Patient Output f(x): **79.0%** | **✓ Additivity Verified: E[f(x)] + Σ ϕ_i = f(x)**

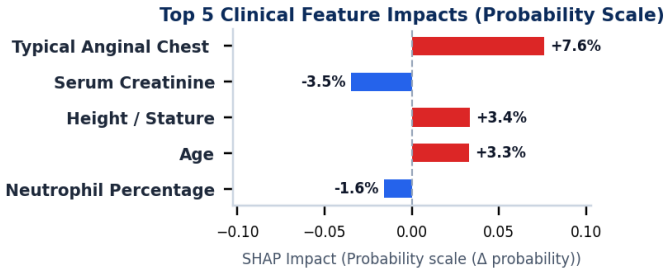
| Rank | Clinical Feature | Category | Patient Value | SHAP (probability) | Impact Direction |
|---|---|---|---|---|---|
| #1 | Typical Anginal Chest Pain | Symptoms | Present (1) | +15.3% | Increases Risk |
| #2 | Regional Wall Motion Abnormality | Echocardiography | Absent (0) | -3.5% | Decreases Risk |
| #3 | Age | Basic Demographics | 58 years | +3.2% | Increases Risk |
| #4 | Atypical Chest Pain | Symptoms | Absent (0) | +3.0% | Increases Risk |
| #5 | Lymphocyte Percentage | Biomarkers | 32 % | +2.5% | Increases Risk |

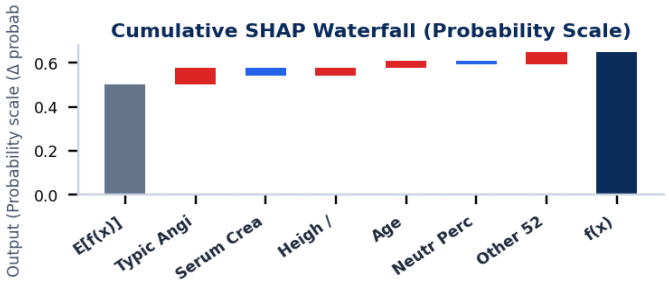



**LCX Artery Model Explainability** (lcx_model.pkl) · Method: **TreeExplainer (Tree SHAP)** · Target: **Stenotic** · **[PROBABILITY SCALE]**
Expected Baseline E[f(x)]: **50.2%** | Final Patient Output f(x): **65.0%** | **✓ Additivity Verified: E[f(x)] + Σ ϕ_i = f(x)**

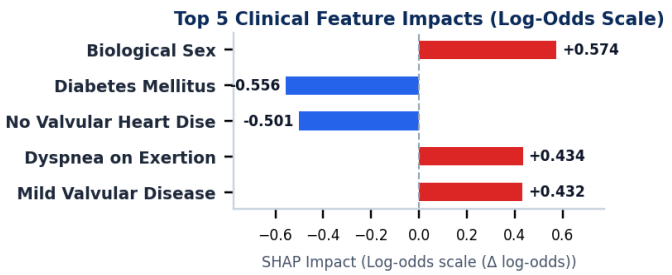
| Rank | Clinical Feature | Category | Patient Value | SHAP (probability) | Impact Direction |
|---|---|---|---|---|---|
| #1 | Typical Anginal Chest Pain | Symptoms | Present (1) | +7.6% | Increases Risk |
| #2 | Serum Creatinine | Biomarkers | 1 mg/dL | -3.5% | Decreases Risk |
| #3 | Height / Stature | Basic Demographics | 170 cm | +3.4% | Increases Risk |
| #4 | Age | Basic Demographics | 58 years | +3.3% | Increases Risk |
| #5 | Neutrophil Percentage | Biomarkers | 60 % | -1.6% | Decreases Risk |

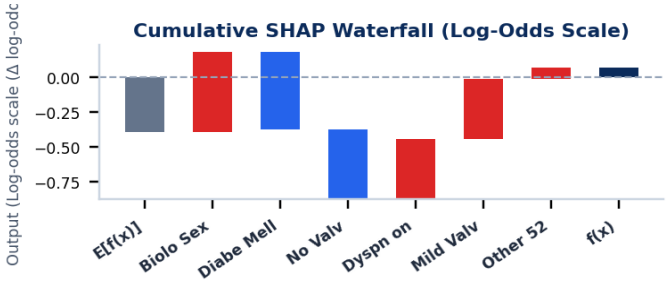



**RCA Artery Model Explainability** (rca_model.pkl) · Method: **LinearExplainer (Linear SHAP)** · Target: **Stenotic** · **[LOG-ODDS SCALE]**
Expected Baseline E[f(x)]: **-0.393 log-odds** | Final Patient Output f(x): **0.068 log-odds** (σ(f(x)) = **51.7%** est. prob) | **✓ Additivity Verified: E[f(x)] + Σ ϕ_i = f(x)**

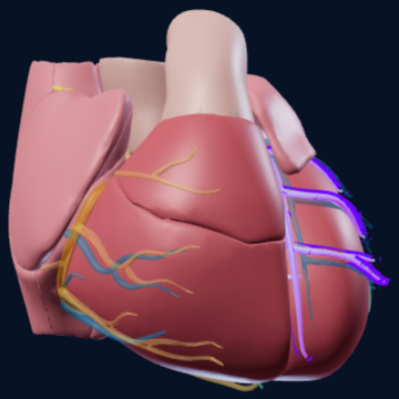
| Rank | Clinical Feature | Category | Patient Value | SHAP (log-odds) | Impact Direction |
|---|---|---|---|---|---|
| #1 | Biological Sex | Basic Demographics | Male | +0.574 log-odds | Increases Risk |
| #2 | Diabetes Mellitus | Medical History | Absent (0) | -0.556 log-odds | Decreases Risk |
| #3 | No Valvular Heart Disease | Valvular Disease | Present (1) | -0.501 log-odds | Decreases Risk |
| #4 | Dyspnea on Exertion | Symptoms | Absent (0) | +0.434 log-odds | Increases Risk |
| #5 | Mild Valvular Disease | Valvular Disease | Absent (0) | +0.432 log-odds | Increases Risk |



## 4. 3D Coronary Heart Visualization

High-resolution anatomical perspective from the interactive Three.js 106-mesh BodyParts3D cardiac model.



**Coronary Risk Telemetry**

Evaluation Mode: **Standard Production Mode**

Spatial Anatomical Risk Overlay

**CAD: 91.7%** (CAD)
**LAD: 79.0%** (Stenotic)
**LCX: 65.0%** (Stenotic)
**RCA: 51.7%** (Stenotic)

**Vascular Color Legend:**

- LAD Arterial Trunk (#2a86ef)
- LCX Circumflex Trunk (#0fb77d)
- RCA Coronary Trunk (#ff9900)
- LM Coronary Stem (#e6d6c2)
- Ventricular Myocardium (#c94a53)

**Anatomical Notice:** 106-mesh BodyParts3D cardiac assembly; generic reference geometry.

## 5. Model Performance and Reliability

**Standard Mode:** Retrospective model evaluation metrics recorded during development on held-out test data.

| Model | Active Model File | Target Vascular Outcome | Accuracy | F1 Score | ROC-AUC |
|---|---|---|---|---|---|
| **CAD Model** | cad_model.pkl | Overall Coronary Artery Disease | **85.2%** | 89.7% | **0.917** |
| **LAD Model** | lad_model.pkl | Left Anterior Descending Stenosis | **65.6%** | 67.7% | **0.756** |
| **LCX Model** | lcx_model.pkl | Left Circumflex Stenosis | **57.4%** | 53.6% | 0.584 |
| **RCA Model** | rca_model.pkl | Right Coronary Artery Stenosis | **60.7%** | 50.0% | 0.652 |

**Model Reliability Notice:** Standard LCX and RCA baseline models exhibit limited discriminatory power in development cohort testing (ROC-AUC 0.584 and 0.652). Probabilities must be interpreted with extreme caution and never guide clinical interventions independently.

## 6. Research Limitations, Educational Guidance & Disclaimer

**Scope of Research Evaluation & Clinical Considerations:**

- Model estimates represent statistical pattern matching against retrospective cohorts, not definitive clinical diagnoses.
- Results must always be interpreted in full clinical context by a licensed medical practitioner.
- Machine learning probabilities cannot substitute for direct coronary angiography, CT angiography, or stress imaging.
- Diagnostic and management decisions depend on patient presentation, physical exam, and comprehensive clinical history.

**RESEARCH PROTOTYPE — NOT A CERTIFIED MEDICAL DEVICE:**

This software and report are intended strictly for educational, scientific research, and methodology evaluation purposes. They are not approved or cleared by any regulatory health agency (e.g., FDA, CE, PMDA) for clinical diagnostic or therapeutic decision-making. Predictions and SHAP explainability analyses do not substitute for formal cardiovascular imaging, laboratory testing, or physician consultation.